



World Health
Organization



Parkinson disease

A public health approach

Technical brief

Parkinson disease

A public health approach

Technical brief

Parkinson disease: a public health approach. Technical brief

ISBN 978-92-4-005098-3 (electronic version)

ISBN 978-92-4-005099-0 (print version)

© World Health Organization, 2022

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; <https://creativecommons.org/licenses/by-nc-sa/3.0/igo>).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: "This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition".

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization (<http://www.wipo.int/amc/en/mediation/rules/>).

Suggested citation. Parkinson disease: a public health approach. Technical brief. Geneva: World Health Organization; 2022. Licence: [CC BY-NC-SA 3.0 IGO](#).

Cataloguing-in-Publication (CIP) data. CIP data are available at <http://apps.who.int/iris>.

Sales, rights and licensing. To purchase WHO publications, see <http://apps.who.int/bookorders>. To submit requests for commercial use and queries on rights and licensing, see <https://www.who.int/copyright>.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

Photos by: cover Pexels/ Hamza Uran; page 4 Unsplash/ Gus Moretta; page 7 Unsplash/ Steven Su; page 8 Unsplash/ Jana Sabeth; page 15 Unsplash/ Greg Rosenke; page 25 Unsplash/ Joshua Hoehne

Layout by Café.art.br

Contents

Acknowledgements	iv
Introduction.....	1
Methods	3
Review of the evidence.....	5
Disease prevention and risk reduction	5
Treatment gaps and challenges to access to care	6
Neurological workforce and treatment	6
Awareness and education.....	8
Opportunities for service delivery and management in health systems	8
Primary care.....	8
Interdisciplinary care	9
Support for carers and self-management	10
Key actions	12
Global health policies to implement PD strategies	12
Advocacy and awareness-raising	12
Prevention and risk reduction	13
Ensuring the availability of basic drugs and therapies.....	13
Strengthening health and social systems and building capacity	14
Carer support.....	15
Research on Parkinson disease	15
References	16

Acknowledgements

Vision and conceptualization

The Parkinson disease technical brief has been developed under the overall guidance and conceptualization of Tarun Dua, Neerja Chowdhary and Nicoline Schiess, WHO Department of Mental Health and Substance Use.

Richard Brown, Chemical Safety and Health Unit, Department of Environment, Climate Change and Health, WHO

Alexandra Rauch, Sensory Functions, Disability and Rehabilitation Unit, Noncommunicable Disease (NCD) Department, WHO

Yuka Sumi, Department of Maternal, Newborn, Child & Adolescent Health & Ageing, WHO

Project coordination and editing

The team that coordinated the development of the PD technical brief, including writing and editing, was comprised of:

Nicoline Schiess, WHO Department of Mental Health and Substance Use

Natasha Fothergill-Misbah, Newcastle University, Newcastle-upon-Tyne, United Kingdom

Rodrigo Cataldi, WHO Department of Mental Health and Substance Use

Neerja Chowdhary, WHO Department of Mental Health and Substance Use

Hiral Shah, Columbia University Medical Center, New York City (NY), United States of America (USA)

Lynda Nwabuobi, Weill Cornell Medical Center, New York City (NY), USA

Technical contribution and review

Valuable material, help and advice was received from technical staff at WHO headquarters and WHO regional and country offices. For their technical guidance and support for the project, we are grateful to:

Expert reviewers

Many international experts including those who participated in the “*Addressing the public health importance of Parkinson disease*” virtual meeting in April 2021 have made vital contributions to the development of the Parkinson disease technical brief. WHO gratefully acknowledges the contribution of the following people for their expert opinion and technical input in the development of this technical brief:

Maria Barretto, Parkinson's Disease and Movement Disorder Society, Mumbai, India; Roongroj Bhidayasiri, Chulalongkorn University Hospital, Bangkok, Thailand; Bastiaan Bloem, Stichting Radboud Universitair Medisch Centrum, Amsterdam, Netherlands; Lorraine Chishimba, University Teaching Hospital, Lusaka, Zambia; Esther Cubo, Hospital Universitario Burgos, Burgos, Spain; Alessandro Di Rocco, Zucker School of Medicine at Hofstra Northwell; Rachel Dolhun, The Michael J. Fox Foundation, New York City (NY), USA; Ray Dorsey, University of Rochester, Rochester (NY), USA; Christopher Dowrick, University of Liverpool, Liverpool, United Kingdom; Natasha Fothergill-Misbah, Newcastle University, Newcastle-upon-Tyne, United Kingdom; Victor Fung, Westmead Hospital and University of Sydney, Sydney,

Australia; Oscar S. Gershanik, Institute of Neuroscience, Favaloro Foundation University Hospital, Buenos Aires, Argentina; Larry Gifford, PD Avengers, Vancouver, Canada; Joyce Gordon, Neurological Health Charities, Canada; Hanan Khalil, Qatar University, College of Health Sciences, Department of Physiotherapy, Doha, Qatar; Andrea Kühn, Charité University Hospital, Berlin, Germany; Sara Lei Heong Lew, Malaysian Parkinson's Disease Association, Kuala Lumpur, Malaysia; Shen-Yang Lim, University of Malaya, Kuala Lumpur, Malaysia; Maria Marano, Parkinson Canada, Toronto, Canada; Jackie Micallef, Parkinson Canada, Toronto, Canada; Jolynne Mokaya, University of Oxford, Oxford, United Kingdom; Emile Moukheiber, Johns Hopkins University School of Medicine, Baltimore (MD), USA; Njideka Okubadejo, College of Medicine, University of Lagos, Lagos, Nigeria; Michael Okun, University of Florida, Gainesville (FL), USA; Pramod Kumar Pal, National Institute of Mental Health and Neurosciences, Bengaluru, India; Hiral Shah, Columbia University Medical Center, New York City (NY), USA; Ali Shalash, Department of Neurology, Ain Shams University, Cairo, Egypt; Todd Sherer, The Michael J. Fox Foundation, New York City (NY), USA; Bernadette Siddiqui, The Michael J. Fox Foundation, New York City (NY), USA; Ted Thompson, Public Policy, The Michael J. Fox Foundation, New York City (NY), USA; Andreas Ullrich, Charité University Hospital, Berlin, Germany; Richard Walker, Northumbria Healthcare NHS Foundation Trust, Newcastle-upon-Tyne, United Kingdom.

Special thanks

We are grateful for the feedback received from those living with Parkinson disease who reviewed this technical brief.

We are grateful for the contribution of the participants of the *“Addressing the public health importance of Parkinson disease”* consultation workshop held 14 and 15 April 2021.

Financial support

WHO gratefully acknowledges the financial support from Edmond J Safra Foundation for the consultation workshop and technical brief.



Introduction

Globally, disability and death due to Parkinson disease (PD) are increasing faster than for any other neurological disorder (1). PD is a degenerative condition of the brain associated with motor symptoms (slow movement, tremor, rigidity, walking and imbalance) and a wide variety of non-motor complications (neuropsychiatric and autonomic symptoms, disorders of sleep and wakefulness, pain and other sensory disturbances). Progression of these symptoms and complications markedly decrease functioning, and quality of life and result in high rates of disability and care requirements (2). The main risk factor for PD is increasing age, although younger people can be affected as well (3). Despite the significant impact of PD, there is global inequality in the availability of neurological resources to manage the disease, with poor availability especially in low- and middle-income countries (LMIC) (4).

The prevalence of PD has doubled in the past 25 years (1). The numbers are likely even higher when the many people living with various forms of parkinsonism are included, such as those caused by degenerative conditions (atypical parkinsonism), vascular lesions in the brain or adverse effects of medications such as neuroleptics. The data on incidence and prevalence are inconsistent, particularly for LMIC (5–7) and for ethnic minorities in high-income countries (HIC) (8–10), because of financial and geographical barriers to health care, underreporting of cases, misdiagnosis, lack of awareness of PD and incorrect perceptions that the decline associated with PD (e.g., slowing down or becoming stooped) is part of “normal” ageing (11–16). The inconsistent data make it difficult to estimate the global impact of PD accurately. Current estimates suggest that, in 2019, PD resulted in 5.8 million disability-adjusted life years, an increase of 81% since 2000, and caused 329,000 deaths, an increase of over 100% since 2000. (17).



**Globally,
disability and
death due
to Parkinson
disease are
increasing
faster than
for any other
neurological
disorder.**

In 2017, PD was estimated to cost US\$ 52 billion per year in the USA (18), which will continue to increase as the incidence and prevalence of PD rise. Large scale, rigorous economic data on the cost of PD in LMIC is lacking (19).

Despite substantial development of drugs to treat PD, none have proven to be effective in slowing progression. Levodopa remains the most effective drug for improving most motor and various non-motor symptoms of PD and for improving functioning and quality of life, although it cannot halt the neurodegenerative process. Data on the potential causes of PD and opportunities for risk reduction are, however, accumulating. The possible causes include environmental (e.g.: exposure to pesticides, solvents and air pollution throughout life) (20) and genetic risk factors (e.g., *LRRK2* polymorphisms or mutations) (21).

Levodopa is on the WHO Model Lists of Essential Medicines (22); however, it is not accessible, available, or affordable everywhere, particularly in LMIC. Although optimal PD care also requires an integrated interdisciplinary approach (3, 23), appropriate treatment, rehabilitation and palliative care provided by interdisciplinary teams with expertise in neurological disorders (4) are lacking in many parts of the world.

In the context of providing universal health coverage, an urgent public health response is necessary to meet the health and social requirements of people with PD and to improve functioning, quality of life and prevent disability as global longevity increases. A pressing need for effective preventive actions is also needed to slow the rising incidence of PD before the burden and costs of treatment overwhelm country health services. In line with the WHO's Intersectoral global action plan on epilepsy and other neurological disorders (24), this brief, targeted to policy-makers, health programme managers and planners, health-care providers, researchers,

people with PD, carers and other stakeholders, outlines the crucial areas for intervention in PD and several areas for action, including:

- global health policies to implement strategies focused on PD
- prevention and reduction of risks for PD including education and awareness;
- assured access to treatment and care, including access to and education of an interdisciplinary workforce trained in neurological disorders; and
- delivery of services and the management of PD at various levels of health systems.

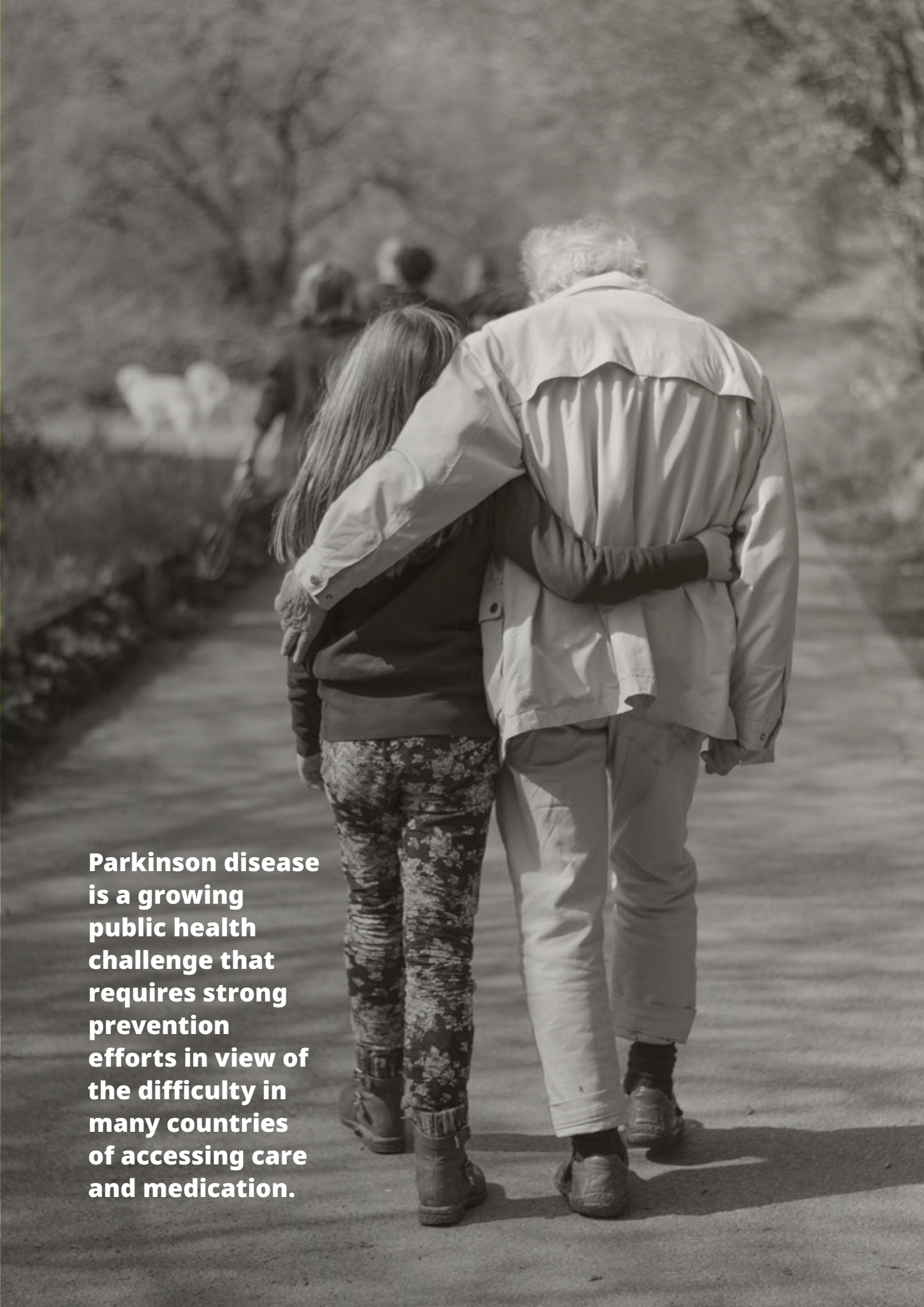


Methods

This technical brief is based on evidence from a WHO-commissioned literature review and a 2-day consultation with international experts on global policy, considerations for implementation and the research agenda for PD, with a focus on LMIC.



Although optimal PD care requires an integrated interdisciplinary approach, appropriate treatment, rehabilitation and palliative care provided by interdisciplinary teams with expertise in neurological disorders are lacking in many parts of the world.



Parkinson disease is a growing public health challenge that requires strong prevention efforts in view of the difficulty in many countries of accessing care and medication.

Review of the evidence

Disease prevention and risk reduction

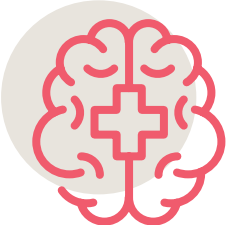
PD is a growing public health challenge that requires strong prevention efforts (25); however, the health-care workforce, policy-makers and the general public are not adequately aware of the importance of reducing risk. Genetic predisposition is the causal factor in only a relatively small sub-group of cases (3), and a number of studies have shown that environmental factors, including pesticides (26–29), air pollution (30, 31) and industrial solvents (32) could increase the risk of PD. Measures used to protect users from exposure to pesticides (e.g., personal protective equipment (PPE), safe application equipment), often are not feasible in LMIC (e.g., too costly or impractical in hot climates) (33) and while less hazardous alternatives often exist and are available, they are not utilized (34). Genetic predisposition may further increase the risk of PD after exposure to environmental toxins (35, 36). As social standing can affect exposure to risk factors, addressing social inequalities would also reduce the development of PD. Some studies suggest that certain lifestyle factors protect against PD, including regular physical exercise (37, 38), a Mediterranean diet (39, 40) and consumption of caffeine (41).



Environmental factors, including pesticides, could increase the risk of Parkinson disease.

Essential health services in an interdisciplinary approach to care can improve functioning and the quality of life of people with PD and reduce the strain on carers. As PD progresses, rehabilitation to improve and maintain functioning and quality of life, to prevent complications and to minimize long-term disability (42–45) and palliative care (46, 47) are considered essential. These interventions are not, however, available everywhere, particularly in LMIC.

Gaps and challenges



23%
of countries
worldwide had
neurologists in
rural areas.

Interdisciplinary care teams, considered to be essential for optimal management of PD, are also lacking in low-resource settings.

NEUROLOGICAL WORKFORCE AND TREATMENT ACCESS

While PD is best diagnosed by specialists with expertise in movement disorders (48), very few such specialists work in LMIC (49, 50) and even fewer in rural areas around the world. The WHO Neurology Atlas provides estimates of 0.03 neurologists per 100 000 population in low-income countries and 4.74 per 100 000 population in HIC, and only 23% of countries worldwide had neurologists in rural areas in 2017 (4). Interdisciplinary care teams, considered to be essential for optimal management of PD (3), are also lacking in low-resource settings (51), where there is limited access to affordable PD nurses or ancillary services such as physiotherapists, occupational therapists, speech and language therapists, and psychologists (7). For example, specialized PD nurses are available in only two countries in Africa and PD speech programmes in only one (52). Clinical diagnosis of PD by trained, non-specialized health-care workers and simplified treatment guidelines could offer better management in primary care where specialist neurological treatment, rehabilitation and palliation provided by interdisciplinary teams are unavailable. Telemedicine might also be used to provide care to people with PD living in underserved areas of the world with geographical and financial barriers to care (53–57).

Government expenditure on health in developing countries is often inadequate, and lack of health insurance coverage means that people with PD must pay out-of-pocket for basic health care and medication (4, 52). The WHO Neurology Atlas reported that

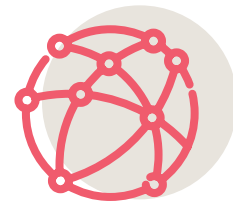
levodopa, which remains the most effective medication for PD (58, 59), was consistently available in primary care in only 34% of the 110 countries surveyed, none of which were low-income countries (4). Limited access to levodopa means that people living with PD cannot be given even the most basic therapy to improve their quality of life. When levodopa is available, it is often unaffordable or is not subsidized by public health-care systems because of poor health insurance coverage (7, 60, 61).

Advanced treatment options for PD, such as deep brain stimulation, can reduce drug intake and costs and improve health and functioning and has been shown to be cost-effective in the long-term (62). Use of deep brain stimulation in LMIC is, however, limited because of the high cost (15) and lack of facilities and expertise (7, 15). Inequitable access and gender disparities are also seen in HIC, where women are much less likely to receive this treatment than men (63, 64). Another treatment for patients with advancing disease in HIC is infusions of dopaminergic medications; however, these are even more costly and thus unavailable for most people with PD (15, 65).

The needs of people with neurological disorders for rehabilitation are also generally not met: only 16% of 105 countries reported specialized neurorehabilitation services and 17% reported neurorehabilitation in general rehabilitation units in 2017 (4). Motor impairments, such as dyskinesias (involuntary movements) and dystonias (painful involuntary muscle contractions) (66) contribute to limitations in speech, mobility and thus, restrictions in many life areas. This underscores the need for a more holistic and interdisciplinary approach, combining the use of medication and non-pharmacological interventions, including interventions for rehabilitation such as yoga (67, 68) and speech or swallowing occupational therapy. Despite increasing recognition of the importance of palliative care services (47), they are lacking in many LMIC. A study published in 2013 reported that only 20 countries in the world (8.5%) have integrated palliative care in health services (69).



16%
of countries
reported specialized
neurorehabilitation
services.



8.5%
of countries
have integrated
palliative care
in health services.





Resources and educational material in local languages are rarely available so many people with Parkinson disease and their families are unaware of the disease.

AWARENESS AND EDUCATION

Globally, the population and also non-specialist health-care professionals are generally poorly informed about PD (16), which can result in erroneous assumptions, such as contributing it to a contagious illness, a form of insanity or a “normal” part of aging (70, 71). Such assumptions can contribute to stigmatization, particularly if the symptoms are obvious, and can delay presentation to health-care services and initiation of treatment, often after several misdiagnoses. Little information about PD, such as resources and educational materials in local languages, is available, so that many people with PD and their families are unaware of the disease and its symptoms and do not seek appropriate care (72). Integrated media and public education campaigns could be conducted to avoid stigmatizing perceptions of PD and to create awareness.

Opportunities for service delivery and management in health systems

As PD is a progressive disease, people living with the condition frequently require long-term care. An interdisciplinary approach will ensure optimal disease management (3, 23, 76–78). Providing holistic care may be daunting in low-resource settings, where limited resources impede building an integrated system of care along a continuum of services that includes diagnosis, treatment, rehabilitation and palliative care.

PRIMARY CARE

Primary care, the entry point to health care in most communities, is the logical setting for addressing PD in LMIC where access to specialist services is lacking. PD is most often diagnosed by clinical evaluation (3), with little or no need for ancillary testing. Diagnosis may therefore be done in primary care facilities by trained health-care workers, including general practitioners and specialist nurses. Complex cases could be triaged to secondary

care for further, specialized assessment, for example, by neurologists, when available. Telemedicine, including sharing of videotaped neurological examinations, is a flexible option and is effective for consulting experts elsewhere and for improving local diagnostic skills (55, 57, 79). Use of telemedicine is in accordance with the WHO Global strategy on digital health 2020–2025 (80). Interventions for rehabilitation, including physiotherapy, occupational, speech and language therapy, diet management and psychological interventions, should be available at all service levels, depending on country context.

Increasing the number of health-care professionals who enrol in specialized neurology training could bridge the gap in access to care for PD, particularly in LMIC.

INTERDISCIPLINARY CARE AND MANAGEMENT

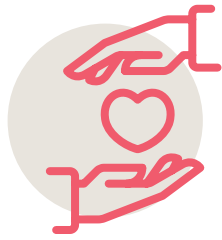
As with many degenerative neurologic disorders, nonpharmacological management can offer symptomatic relief of motor or non-motor symptoms in PD. Several studies have investigated specific forms and approaches to physiotherapy including strength training, gait and balance training, intense rehabilitation training and hydrotherapy (23, 76, 77, 78). Occupational therapy can help people with PD better manage their problems in daily functioning such as swallow therapies which are important adjunctive treatments in PD and are especially important in preventing complications of pulmonary aspiration.

Planning for palliative care might have to start at diagnosis (46) and not just at crises, although such discussions may be difficult (83–85). Little attention has been paid to the provision of palliative care and therefore improvements in the quality of life of people with PD (47, 86). Palliative services are not common in LMIC. Primary health-care workers with appropriate understanding of PD could be given the relevant training to discuss the benefits of advance planning of palliative care with people with PD and their carers including due consideration of local and cultural perspectives about the end of life. Development of resources and programmes that can be delivered by carers of people with advancing PD could also overcome limited access to palliative care (87, 88).



Diagnosis of Parkinson disease can be done in primary care facilities by trained health-care workers.





Effective delivery of care for Parkinson disease should include support for carers.

SUPPORT FOR CARERS AND SELF-MANAGEMENT

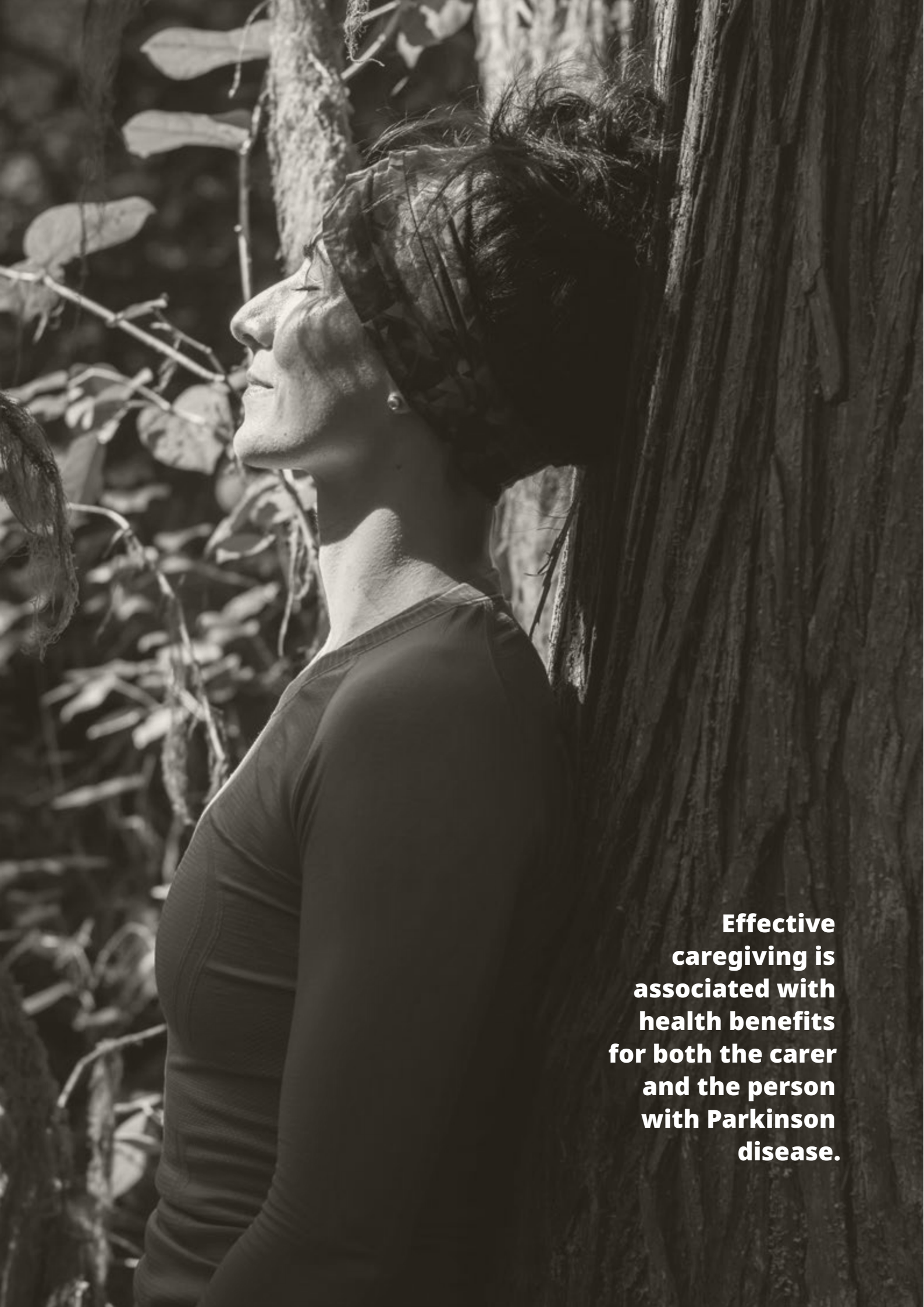
Self-management is an important component of care for PD, as progression of the disease requires management of motor and non-motor symptoms, such as lack of balance with a propensity to falls and cognitive and behavioural problems. These symptoms in people with advanced disease add to the stress and burden of carers (81). Effective delivery of care for PD should therefore include support for carers (82). Support groups of people with PD (both in-person and virtual) are beneficial for learning and communicating and provide emotional and social support for both people with PD and their families. Support groups also offer opportunities for local advocacy, combatting stigmatization in communities and educating people about PD. They may also contribute to global advocacy and awareness (72).

Models of care, educational programmes and advocacy in LMIC

Examples of community care, educational programmes and advocacy initiatives in low-resource settings (73) have shown that access to treatment and awareness can be improved.

In India, a community model of care developed in 2005 by the Parkinson's Disease and Movement Disorder Society, has helped overcome barriers to access to and affordability of health care by using and building community resources. The interdisciplinary model is delivered in 65 support centres and has served over 50 000 people with PD and their families, promoting self-management to improve independence and quality of life and to reduce the burden and stress of carers (74).

In Uruguay, a therapeutic, rehabilitative and educational programme was established in 2007 to bridge gaps in access to PD care (75). Professionals deliver rehabilitation and provide mental health support and counselling to people with PD and their families and carers. Medical professionals can also attend educational lectures. By the end of 2018, the programme had reached 423 people with PD and 431 family members. These initiatives demonstrate the importance of the community in the management of PD.



**Effective
caregiving is
associated with
health benefits
for both the carer
and the person
with Parkinson
disease.**

Key actions

As the prevalence of PD is increasing globally, it is a challenge to many populations. Political commitment will be required to develop strategies, programmes and services that are effective for people with PD and their families and carers. The key actions identified in this brief are: global health policies to implement PD strategies; advocacy and awareness-raising; prevention and risk reduction; ensuring the availability of basic drugs and interdisciplinary therapies such as rehabilitation and palliative care; strengthening health and social systems and building capacity; and research on PD.

Global health policies to implement PD strategies

- leverage the momentum of the Intersectoral global action plan on epilepsy and other neurological disorders to advance the agenda of PD in all settings;
- include civil society organizations, people with PD and their support networks to increase the success and impact of PD strategies.

Advocacy and awareness-raising

- facilitate global action to change local, national and international policy to improve care and services for PD;
- improve awareness among people with PD, their families and society (with local, population-level solutions) to increase understanding of PD and to change public attitudes and practices, particularly stigmatization and discrimination; and
- promote the establishment of local PD support groups and community initiatives.



Prevention and risk reduction

- ban pesticides (e.g.: paraquat and chlorpyrifos) and chemicals (e.g.: trichloroethylene) which have been linked to PD and develop safer alternatives as per WHO guidance (89–91);
- accelerate action to reduce levels of and exposure to air pollution, an important risk factor for PD and other noncommunicable diseases, including Alzheimer disease, respiratory disease, heart disease and stroke;
- further understand and address modifiable risk factors through public education and changes to policy and legislation;
- increase access to and promotion of healthy behaviour (i.e., exercise and diet) along the life-course to reduce the risk for PD and to enhance potentially protective factors;
- encourage links for prevention of PD with promotion of healthy living, working and environmental conditions; and
- identify synergies between PD, other neurological disorders and noncommunicable diseases to facilitate prevention.

Ensuring the availability of medicines

- increase access to and the affordability of therapies for PD of any severity, including motor and non-motor features;
- promote rational selection of medicines for training, supply and reimbursement including updates of national essential medicine lists;
- adapt/modify local legislation to enable the production and increase access to generic PD medicines;
- promote and support local manufacture of PD drugs and therapies in LMIC;
- generate evidence to feed into the national assessment of procurement and supply chain management;
- inform and support national planners/ partners to prepare risk management plans for managing essential health product supply chains and for maintaining equitable availability, access, and appropriate distribution of medicines for PD;
- register PD medicines (e.g., levodopa) with relevant pharmaceutical registration boards in LMIC to improve its availability;
- review and revise the WHO Essential Medicines list to include dopaminergic and other medications (e.g., amantadine); and
- negotiate with pharmaceutical and device companies to ensure that treatments are affordable for more people with PD.



Political commitment will be required to develop strategies, programmes and services that are effective for people with Parkinson disease.

Strengthening health and social systems and building capacity

- build the capacity of primary healthcare, social workers, rehabilitation and palliative care;
- develop career tracks for the neurological workforce by strengthening postgraduate training and working in partnership with medical societies to raise awareness of PD;
- promote educational programmes within the PD interdisciplinary health and rehabilitation workforce and create networks for referrals;
- provide education and training of primary health-care providers, including allied health professionals, and shift tasks to community settings;
- strengthen information systems for better recording of the numbers of people affected by PD globally, particularly in low-resource settings and in various population groups;
- provide educational materials for better diagnosis, communicating the diagnosis and follow-up care, including support;
- increase innovations for specialist support (e.g.: telemedicine) to offer remote interdisciplinary support for people with PD, with respect for their privacy and security (in accordance with the WHO Global strategy on digital health 2020–2025);
- integrate diagnosis, treatment, rehabilitation and palliative services into interdisciplinary teams within universal health coverage;
- ensure good health financing and social protection to prevent catastrophic out-of-pocket payments and early retirement.

Carer support

- increase awareness of the impact of caring for people with PD, including the need to protect carers from discrimination, support their ability to continue to provide care throughout the disease progression, and promote self-advocacy;
- provide accessible and evidence-based information on available resources in the community, such as training programmes, respite care, mental health services and other resources tailored to the needs of carers of people with PD;
- assist in implementing culturally sensitive, context-specific and person-centred training programmes for carers and families to promote well-being and enhance knowledge and caregiving skills throughout the progression of PD, starting with existing resources such as WHO's iSupport (92) and mhGAP (93).

Research on Parkinson disease

- collect robust, standardized, easily accessible data, specifically prevalence, incidence and risk factors in representative populations, from routine health information systems;
 - develop better regulatory frameworks to allow more efficient and secure sharing of data between sectors and across countries and regions;
 - improve coordination of multi-stakeholder, multi-centre, cross-regional research collaborations and to build research capacity, particularly in LMIC;
 - improve representation of LMIC in PD research through cost-effective, locally relevant strategies;
 - promote diversity, equity and inclusion in all areas of PD research and care;
 - involve and support people with PD, their families and carers by active participation in and guidance of research on innovative solutions with a meaningful impact;
 - evaluate the feasibility and effectiveness of community models of care for people with PD; and
 - support not-for-profit organizations financially in the design, implementation and documentation of action research to build models of care that can be replicated.
- promote and extend global funding for PD research;
 - generate additional knowledge and evidence to inform policy, planning and programming for PD;
 - increase investment in biomedical, clinical, implementation and translational research into new diagnostics and treatments in direct response to the needs of people with PD;

References

1. GBD 2016 Neurology Collaborators. Global, regional, and national burden of neurological disorders, 1990–2016: A systematic analysis for the global burden of disease study 2016. *Lancet Neurol.* 2019;18(5):459–480. [https://www.thelancet.com/journals/lanneur/article/PIIS1474-4422\(18\)30499-X/fulltext](https://www.thelancet.com/journals/lanneur/article/PIIS1474-4422(18)30499-X/fulltext)
2. Shulman LM. Understanding disability in Parkinson's disease. *Movement Disord.* 2010;25(Suppl 1):S131–5. <https://movementdisorders.onlinelibrary.wiley.com/doi/abs/10.1002/mds.22789>
3. Bloem BR, Okun MS, Klein C. Parkinson's disease. *Lancet.* 2021;397(10291):2284–303. [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(21\)00218-X/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(21)00218-X/fulltext)
4. Atlas: country resources for neurological disorders, second ed. Geneva: World Health Organization; 2017. <https://apps.who.int/iris/handle/10665/258947>
5. Okubadejo NU, Bower JH, Rocca WA, Maraganore DM. Parkinson's disease in Africa: A systematic review of epidemiologic and genetic studies. *Movement Dis.* 2006;21(12):2150–6. <https://movementdisorders.onlinelibrary.wiley.com/doi/abs/10.1002/mds.21153>
6. Akinyemi RO. Epidemiology of Parkinsonism and Parkinson's disease in sub-Saharan Africa: Nigerian profile. *J Neurosci in Rural Pract.* 2012;3(3):233–4. <https://www.thieme-connect.de/products/ejournals/pdf/10.4103/0976-3147.102586.pdf>
7. Williams U, Bandmann O, Walker R. Parkinson's disease in sub-Saharan Africa: A review of epidemiology, genetics and access to care. *J Movement Disord.* 2018;11(2):53–64. <https://www.e-jmd.org/upload/jmd-17028.pdf>
8. van den Eeden SK, Tanner CM, Bernstein AL, Fross RD, Leimpeter A, Bloch DA et al. Incidence of Parkinson's disease: Variation by age, gender, and race/ethnicity. *Am J Epidemiol.* 2003;157(11):1015–22. <https://academic.oup.com/aje/article/157/11/1015/151509>
9. Di Luca DG, Sambursky JA, Margolesky J, Cordeiro JG, Diaz A, Shpiner DS et al. Minority enrollment in Parkinson's disease clinical trials: Meta-analysis and systematic review of studies evaluating treatment of neuropsychiatric symptoms. *J Parkinsons Dis.* 2020;10(4):1709–16. <https://pubmed.ncbi.nlm.nih.gov/32894250/>
10. Hemming JP, Gruber-Baldini AL, Anderson KE, Fishman PS, Reich SG, Weiner WJ et al. Racial and socioeconomic disparities in parkinsonism. *Arch Neurol.* 2011;68(4):498–503. <https://pubmed.ncbi.nlm.nih.gov/21149802/>
11. Dahodwala N, Karlawish J, Siderowf A, Duda JE, Mandell DS. Delayed Parkinson's disease diagnosis among African-Americans: The role of reporting of disability. *Neuroepidemiology.* 2011;36(3):150–4. <https://www.karger.com/Article/Abstract/324935>
12. Dahodwala N, Siderowf A, Xie M, Noll E, Stern M, Mandell DS. Racial differences in the diagnosis of Parkinson's disease. *Movement Disord.* 2009;24(8):1200–5. <https://movementdisorders.onlinelibrary.wiley.com/doi/10.1002/mds.22557>
13. Pan S, Stutzbach J, Reichwein S, Lee BK, Dahodwala N. Knowledge and attitudes about Parkinson's disease among a diverse group of older adults. *J Cross-cultural Gerontol.* 2014;29(3):339–52. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4134410/>
14. Wright Willis A, Evanoff BA, Lian M, Criswell SR, Racette BA. Geographic and ethnic variation in Parkinson disease: A population-based study of US Medicare beneficiaries. *Neuroepidemiology.* 2010;34(3):143–51. <https://pubmed.ncbi.nlm.nih.gov/20090375/>
15. Lim SY, Tan AH, Ahmad-Annuar A, Klein C, Tan LCS, Rosales RL et al. Parkinson's disease in the Western Pacific Region. *Lancet Neurol.* 2019;18(9):865–79. <https://pubmed.ncbi.nlm.nih.gov/31175000/>

16. Fothergill-Misbah N, Walker R, Kwasa J, Hooker J, Hampshire K. "Old people problems", uncertainty and legitimacy: Challenges with diagnosing Parkinson's disease in Kenya. *Soc Sci Med*. 2021;282:114148. <https://pubmed.ncbi.nlm.nih.gov/34153822/>
17. Global health estimates 2020: Disease burden by cause, age, sex by country and by region, 2000–2019. Geneva: World Health Organization; 2020. <https://apps.who.int/iris/bitstream/handle/10665/332070/9789240005105-eng.pdf>
18. Yang W, Hamilton JL, Kopil C, Beck JC, Tanner CM, Albin RL et al. Current and projected future economic burden of Parkinson's disease in the US. *NPJ Parkinsons Dis*. 2020;6:15. <https://pubmed.ncbi.nlm.nih.gov/32665974/>
19. Bovolenta TM, de Azevedo Silva SM, Arb Saba R, Borges V, Ferraz HB, Felicio AC. Systematic review and critical analysis of cost studies associated with parkinson's disease. *Parkinsons Dis*. 2017;2017:3410946. <https://pubmed.ncbi.nlm.nih.gov/28357150/>
20. Ascherio A, Schwarzschild MA. The epidemiology of Parkinson's disease: risk factors and prevention. *Lancet Neurol*. 2016;15(12):1257–72. <https://pubmed.ncbi.nlm.nih.gov/27751556/>
21. Deng H, Wang P, Jankovic J. The genetics of Parkinson disease. *Ageing Res Rev*. 2018;42:72–85. <https://www.sciencedirect.com/science/article/pii/S1568163717302453>
22. WHO Model Lists of Essential Medicines. <https://www.who.int/groups/expert-committee-on-selection-and-use-of-essential-medicines/essential-medicines-lists>. Last updated: 2021. Accessed: 02/05/2022
23. Radder DLM, de Vries NM, Riksen NP, Diamond SJ, Gross D, Gold DR et al. Multidisciplinary care for people with Parkinson's disease: The new kids on the block! *Expert Rev Neurotherapeutics*. 2019;19(2):145–57. <https://www.tandfonline.com/doi/s/10.1080/14737175.2019.1561285?journalCode=iern20>
24. Draft Intersectoral global action plan on epilepsy and other neurological disorders 2022-2031. <https://www.who.int/news/item/12-01-2022-draft-intersectoral-global-action-plan-on-epilepsy-and-other-neurological-disorders-2022-2031>. Updated: 12/01/2022. Accessed: 02/05/2022
25. Neurological disorders: Public health challenges. Geneva: World Health Organization; 2006. <https://apps.who.int/iris/handle/10665/43605>
26. Gunnarsson LG, Bodin L. Occupational exposures and neurodegenerative diseases. A systematic literature review and meta-analyses. *Int J Environ Res Public Health*. 2019;16(3):337. <https://pubmed.ncbi.nlm.nih.gov/30691095/>
27. Tangamornsuksan W, Lohitnavy O, Sruamsiri R, Chaiyakunapruk N, Norman Scholfield C, Reisfeld B et al. Paraquat exposure and Parkinson's disease: A systematic review and meta-analysis. *Arch Environ Occup Health*. 2019;74(5):225–38. <https://pubmed.ncbi.nlm.nih.gov/30474499/>
28. Vaccari C, Dib R, Gomaa H, Lopes L, De Camargo JL. Paraquat and Parkinson's disease: A systematic review and meta-analysis of observational studies. *Jo Toxicol Environ Health B*. 2019;22:1–31. <https://pubmed.ncbi.nlm.nih.gov/31476981/>
29. Ascherio A, Chen H, Weisskopf MG, O'Reilly E, McCullough ML, Calle EE et al. Pesticide exposure and risk for Parkinson's disease. *Ann Neurol*. 2006;60(2):197–203. <https://www.semanticscholar.org/paper/Pesticide-exposure-and-risk-for-Parkinson's-disease-Ascherio-Chen/026d6666e943c8c996e9c47d6b81dee3a64ad994>
30. Ritz B, Lee PC, Hansen J, Lassen CF, Ketznel M, Sørensen M et al. Traffic-related air pollution and Parkinson's disease in Denmark: A case-control study. *Environ Health Perspect*. 2016;124(3):351–6. <https://www.semanticscholar.org/paper/Traffic-Related-Air-Pollution-and-Parkinson%E2%80%99s-in-A-Ritz-Lee/765953e5c123cc0efc6151e54042dc46677770f9>
31. Jo S, Kim YJ, Park KW, Hwang YS, Lee SH, Kim BJ, et al. Association of NO₂ and other air pollution exposures with the risk of Parkinson disease. *JAMA Neurol*. 2021;78(7):800–8. <https://europepmc.org/article/MED/33999109>
32. Goldman SM, Quinlan PJ, Ross GW, Marras C, Meng C, Bhudhikanok GS et al. Solvent exposures and Parkinson disease risk in twins. *Ann Neurol*. 2012;71(6):776–84. <https://europepmc.org/article/MED/22083847>

33. FAO and WHO. 2020. Guidelines for personal protection when handling and applying pesticide – International Code of Conduct on Pesticide Management. Rome. <https://www.fao.org/3/ca7430en/CA7430EN.pdf>
34. FAO and WHO. International code of conduct on pesticide management: guidelines on highly hazardous pesticides. 2016. Available at: <https://www.who.int/publications/i/item/9789241510417>
35. Warner TT, Schapira AHV. Genetic and environmental factors in the cause of Parkinson's disease. *Ann Neurol*. 2003;53(S3):S16–25. <https://www.semanticscholar.org/paper/Genetic-and-environmental-factors-in-the-cause-of-Warner-Schapira/b5950776906190aac07b0d34b397596a2a319d51>
36. Dardiotis E, Xiomerisiou G, Hadjichristodoulou C, Tsatsakis AM, Wilks MF, Hadjigeorgiou GM. The interplay between environmental and genetic factors in Parkinson's disease susceptibility: The evidence for pesticides. *Toxicology*. 2013;307:17–23. <https://www.semanticscholar.org/paper/The-interplay-between-environmental-and-genetic-in-Dardiotis-Xiomerisiou/d35f77691c1b674d122c5cab6cc325e8a4848c82>
37. van der Kolk NM, de Vries NM, Kessels RPC, Joosten H, Zwinderman AH, Post B et al. Effectiveness of home-based and remotely supervised aerobic exercise in Parkinson's disease: A double-blind, randomised controlled trial. *Lancet Neurol*. 2019;18(11):998–1008. <https://pubmed.ncbi.nlm.nih.gov/31521532/>
38. Olsson TT, Svensson M, Hållmarker U, James S, Deierborg T. Delayed clinical manifestation of Parkinson's disease among physically active: Do participants in a long-distance ski race have a motor reserve? *J Parkinsons Dis*. 2020;10(1):267–74. <https://pubmed.ncbi.nlm.nih.gov/32623406/>
39. Agarwal P, Wang Y, Buchman AS, Holland TM, Bennett DA, Morris MC. MIND diet associated with reduced incidence and delayed progression of Parkinsonism in old age. *J Nutr Health Aging*. 2018;22(10):1211–5. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6436549/>
40. Maraki MI, Yannakoulia M, Stamelou M, Stefanis L, Xiomerisiou G, Kosmidis MH et al. Mediterranean diet adherence is related to reduced probability of prodromal Parkinson's disease. *Movement Disord*. 2019;34(1):48–57. <https://pubmed.ncbi.nlm.nih.gov/30306634/>
41. Kachroo A, Irizarry MC, Schwarzschild MA. Caffeine protects against combined paraquat and maneb-induced dopaminergic neuron degeneration. *Exp Neurol*. 2010;223(2):657–61. <https://pubmed.ncbi.nlm.nih.gov/20188092/>
42. Bloem BR, de Vries NM, Ebersbach G. Nonpharmacological treatments for patients with Parkinson's disease. *Movement Disord*. 2015;30(11):1504–20. <https://pubmed.ncbi.nlm.nih.gov/26274930/>
43. Abbruzzese G, Marchese R, Avanzino L, Pelosin E. Rehabilitation for Parkinson's disease: Current outlook and future challenges. *Parkinsonism Relat Disord*. 2016;22:S60–64. <https://pubmed.ncbi.nlm.nih.gov/26360239/>
44. Díez-Cirarda M, Ibarretxe-Bilbao N, Peña J, Ojeda N. Neurorehabilitation in Parkinson's disease: A critical review of cognitive rehabilitation effects on cognition and brain. *Neural Plasticity*. 2018;2018:2651918. <https://pubmed.ncbi.nlm.nih.gov/29853840/>
45. Alzahrani H, Venneri A. Cognitive rehabilitation in Parkinson's disease: A systematic review. *J Parkinsons Dis*. 2018;8(2):233–45. <https://pubmed.ncbi.nlm.nih.gov/29614698/>
46. Grimes D, Fitzpatrick M, Gordon J, Miyasaki J, Fon EA, Schlossmacher M et al. Canadian guideline for Parkinson disease. *CMAJ*. 2019;191(36):E989–1004. <https://pubmed.ncbi.nlm.nih.gov/31501181/>
47. Miyasaki JM, Lim SY, Chaudhuri KR, Antonini A, Piemonte M, Richfield E et al. Access and attitudes toward palliative care among movement disorders clinicians. *Movement Disord*. 2021. <https://repositorio.ul.pt/handle/10451/49538?locale=en>
48. Willis AW, Schootman M, Tran R, Kung N, Evanoff BA, Perlmuter JS et al. Neurologist-associated reduction in PD-related hospitalizations and health care expenditures. *Neurology*. 2012;79(17):1774–80. <https://profiles.wustl.edu/en/publications/neurologist-associated-reduction-in-pd-related-hospitalizations-a>
49. Willis AW, Schootman M, Evanoff BA, Perlmuter JS, Racette BA. Neurologist care in Parkinson disease. A utilization, outcomes, and survival study. *Neurology*. 2011;77(9):851–7. <https://pubmed.ncbi.nlm.nih.gov/21832214/>

50. Bhidayasiri R, Virameteekul S, Sukoandari B, Tran TN, Lim TT. Challenges of Parkinson's disease care in Southeast Asia. In: Riederer P, Laux G, Nagatsu T, Le W, Riederer C, editors. *NeuroPsychopharmacotherapy*. Cham: Springer International Publishing; 2020:1–21. <https://www.semanticscholar.org/paper/Challenges-of-Parkinson%E2%80%99s-Disease-Care-in-Southeast-Bhidayasiri-Virameteekul/72a288bf271bd6833026811d74e2d61b0de8c22c>
51. Jourdain VA, Schechtmann G. Health economics and surgical treatment for Parkinson's disease in a world perspective: Results from an international survey. *Stereotactic Function Neurosurg*. 2014;92(2):71–9. <https://www.semanticscholar.org/paper/Health-Economics-and-Surgical-Treatment-for-Disease-Jourdain-Schechtmann/e8d0343b11be1969ebf156376fb4bed9ed20ce2d>
52. Hamid E, Ayele BA, Massi DG, Ben Sassi S, Tibar H, Djonga EE et al. Availability of therapies and services for Parkinson's disease in Africa: A continent-wide survey. *Movement Disord*. 2021. <https://www.semanticscholar.org/paper/Availability-of-Therapies-and-Services-for-Disease-Hamid-Ayele/ce2b138f28b849930ce77eed6e3b25dd36ee8b2e>
53. Cubo E, Doumbe J, López E, Lopez GA, Gatto E, Persi G et al. Telemedicine enables broader access to movement disorders curricula for medical students. *Tremor Other Hyperkinet Mov (N Y)*. 2017;7:501. <https://europepmc.org/article/PMC/5673687>
54. Cubo E, Doumbe J, Njiengwe E, Onana P, Garoña R, Alcalde J et al. A Parkinson's disease tele-education program for health care providers in Cameroon. *J Neurol Sci*. 2015;357(1–2):285–7. <https://www.semanticscholar.org/paper/A-Parkinson%E2%80%99s-disease-tele-education-program-for-in-Cubo-Doumb%C3%A9/946959562137abb58c91537dd0a8514bb9d796de>
55. Shalash A, Spindler M, Cubo E. Global perspective on telemedicine for Parkinson's disease. *J Parkinsons Dis*. 2021;11:S11–8. <https://pubmed.ncbi.nlm.nih.gov/33579872/>
56. Marzinzik F, Wahl M, Doletschek CM, Jugel C, Rewitzer C, Klostermann F. Evaluation of a telemedical care programme for patients with Parkinson's disease. *J Telemed Telecare*. 2012;18(6):322–7. <https://www.semanticscholar.org/paper/Evaluation-of-a-telemedical-care-programme-for-with-Marzinzik-Wahl/3b88da9a4d848f9fdd1f5a1018c6fce77cc8e95d>
57. Schneider RB, Biglan KM. The promise of telemedicine for chronic neurological disorders: The example of Parkinson's disease. *Lancet Neurol*. 2017;16(7):541–51. <https://pubmed.ncbi.nlm.nih.gov/28566190/>
58. Connolly BS, Lang AE. Pharmacological treatment of Parkinson disease: A review. *JAMA*. 2014;311(16):1670–83. <https://pubmed.ncbi.nlm.nih.gov/24756517/>
59. Parkinson's disease in adults: NICE guideline. London: National Institute for Health and Care Excellence; 2017. <https://www.nice.org.uk/guidance/ng71>
60. Mokaya J, Dotchin CL, Gray WK, Hooker J, Walker RW. The accessibility of Parkinson's disease medication in Kenya: Results of a national survey. *Movement Disord Clin Pract*. 2016:376–81. <https://movementdisorders.onlinelibrary.wiley.com/doi/epdf/10.1002/mdc3.12294>
61. Okubadejo NU, Ojo OO, Wahab KW, Abubakar SA, Obiabo OY, Salawu FK et al. A nationwide survey of Parkinson's disease medicines availability and affordability in Nigeria. *Movement Disord Clin Pract*. 2019;6(1):27–33. <https://www.semanticscholar.org/paper/A-Nationwide-Survey-of-Parkinson%E2%80%99s-Disease-and-in-Okubadejo-Ojo/68b87906eb0dcd60f8f9939c0f2502eff7891cb1>
62. Dams J, Siebert U, Bornschein B, Volkmann J, Deuschl G, Oertel WH et al. Cost-effectiveness of deep brain stimulation in patients with Parkinson's disease. *Movement Disord*. 2013;28(6):763–71. <https://www.semanticscholar.org/paper/Cost%E2%80%99effectiveness-of-deep-brain-stimulation-in-Dams-Siebert/f63e0555161e29eef17c974ac930bf34e7b26f33>
63. Chan AK, McGovern RA, Brown LT, Sheehy JP, Zacharia BE, Mikell CB et al. Disparities in access to deep brain stimulation surgery for Parkinson disease: Interaction between African American race and Medicaid use. *JAMA Neurol*. 2014;71(3):291–9. <https://pennstate.pure.elsevier.com/en/publications/disparities-in-access-to-deep-brain-stimulation-surgery-for-parki>

64. Hariz GM, Nakajima T, Limousin P, Foltynie T, Zrinzo L, Jahanshahi M et al. Gender distribution of patients with Parkinson's disease treated with subthalamic deep brain stimulation; a review of the 2000–2009 literature. *Parkinsonism Rel Disord.* 2011;17(3):146–9. <https://www.semanticscholar.org/paper/Gender-distribution-of-patients-with-Parkinson%27s-a-Hariz-Nakajima/441fd7bfe6ea8ab11fe067754be39e17b66845ad>
65. Poewe W, Wenning GK. Apomorphine: An underutilized therapy for Parkinson's disease. *Movement Disord.* 2000;15(5):789–94. <https://www.semanticscholar.org/paper/Apomorphine%3A-An-underutilized-therapy-for-disease-Poewe-Wenning/05c5d6c571adf6364a7ec140e952ce719ee063ad>
66. Marsden CD. Problems with long-term levodopa therapy for Parkinson's disease. *Clin Neuropharmacol.* 1994;17(Suppl 2):S32–44. <https://pubmed.ncbi.nlm.nih.gov/9358193/>
67. Ghaffari BD, Kluger B. Mechanisms for alternative treatments in Parkinson's disease: Acupuncture, tai chi, and other treatments. *Curr Neurol Neurosci Rep.* 2014;14(6):451. <https://pubmed.ncbi.nlm.nih.gov/24760476/>
68. Nijkrake MJ, Keus SH, Kalf JG, Sturkenboom IH, Munneke M, Kappelle AC et al. Allied health care interventions and complementary therapies in Parkinson's disease. *Parkinsonism Rel Disord.* 2007;13(Suppl 3):S488–94. <https://www.semanticscholar.org/paper/Allied-health-care-interventions-and-complementary-Nijkrake-Keus/501246e070fb0716f4dbf7deb6ee5b5e0fb353b5>
69. Lynch T, Connor S, Clark D. Mapping levels of palliative care development: A global update. *J Pain Symptom Manage.* 2013;45(6):1094–106. <https://pubmed.ncbi.nlm.nih.gov/23017628/>
70. Jagota P, Jongsuntisuk P, Plengsri R, Chokpatcharavate M, Phokaewvarangkul O, Chirapravati V et al. If your patients were too embarrassed to go out in public, what would you do? Public education to break the stigma on Parkinson's disease using integrated media. *Patient Rel Outcome Measures.* 2020;12(11):132–48. <https://www.semanticscholar.org/paper/If-Your-Patients-Were-Too-Embarrassed-to-Go-Out-in-Jagota-Jongsuntisuk/fa18c4dd844aa3066ad93c5751981c1b89fc3962>
71. Kaddumukasa M, Kakooza A, Kaddumukasa MN, Ddumba E, Mugenyi L, Sajatovic M et al. Knowledge and attitudes of Parkinson's disease in rural and urban Mukono District, Uganda: A cross-sectional, community-based study. *Parkinsons Dis.* 2015;2015(196150):1–7. <https://www.semanticscholar.org/paper/Knowledge-and-Attitudes-of-Parkinson's-Disease-in-A-Kaddumukasa-Kakooza/5d2c01cfe417eb89c311cbd3460af57ba95f982a>
72. Fothergill-Misbah N, Moffatt S, Mwithiga H, Hampshire K, Walker R. The role of support groups in the management of Parkinson's disease in Kenya: Sociality, information and legitimacy. *Glob Public Health.* 2021:1–11. <https://www.semanticscholar.org/paper/The-role-of-support-groups-in-the-management-of-in-Fothergill-Misbah-Moffatt/8aeda6092883c0b5c967e1056cb9170f2970471>
73. Gourie-Devi M. Organization of neurology services in India: Unmet needs and the way forward. *Neurol India.* 2008;56(1):4–12. <https://pubmed.ncbi.nlm.nih.gov/18310829/>
74. PDMDS model of care. Mumbai: Parkinson's Disease and Movement Disorder Society; 2020. <https://www.parkinsonssocietyindia.com/pdmds-model-of-care/>
75. Martinez Lupi FL. El cuidado en la Enfermedad de Parkinson: revisión nacional e internacional [Care in Parkinson's disease: national and international review]. Montevideo: Faculty of Psychology, University of the Republic of Uruguay; 2018. <https://www.colibri.udelar.edu.uy/jspui/bitstream/20.500.12008/21032/1/Martinez.pdf>
76. Lim SY, Tan AH, Fox SH, Evans AH, Low SC. Integrating patient concerns into Parkinson's disease management. *Curr Neurol Neurosci Rep.* 2017;17(1):3. <https://pubmed.ncbi.nlm.nih.gov/28102483/>
77. Qamar MA, Harington G, Trump S, Johnson J, Roberts F, Frost E. Multidisciplinary care in Parkinson's disease. *Int Rev Neurobiol.* 2017;132:511–23. <https://www.semanticscholar.org/paper/Multidisciplinary-care-for-patients-with-disease.-Marck-Kalf/cef35d6bca70227575d760b18dce52b081f9e3cf>

78. Bloem BR, Henderson EJ, Dorsey ER, Okun MS, Okubadejo N, Chan P et al. Integrated and patient-centred management of Parkinson's disease: A network model for reshaping chronic neurological care. *Lancet Neurol.* 2020;19(7):623–34. <https://www.semanticscholar.org/paper/Integrated-and-patient-centred-management-of-a-for-Bloem-Henderson/394309d2dc7e55b9f4bed0809f8f3b1af4afa6fc>
79. van den Bergh R, Bloem BR, Meinders MJ, Evers LJW. The state of telemedicine for persons with Parkinson's disease. *Curr Opin Neurol.* 2021;34(4):589–97. <https://www.semanticscholar.org/paper/The-state-of-telemedicine-for-persons-with-disease-Bergh-Bloem/3af8499f441b2be00f544061f8efbcf00c12d177>
80. Global strategy on digital health 2020–2025. Geneva: World Health Organization; 2021. <https://apps.who.int/iris/bitstream/handle/10665/344249/9789240020924-eng.pdf>
81. Hand A, Oates LL, Gray WK, Walker RW. The role and profile of the informal carer in meeting the needs of people with advancing Parkinson's disease. *Aging Ment Health.* 2019;23(3):337–44. <https://www.semanticscholar.org/paper/The-role-and-profile-of-the-informal-carer-in-the-Hand-Oates/45371f42e9b8b962681707edc91fbb84a2c57d58>
82. Mosley PE, Moodie R, Dissanayaka N. Caregiver burden in Parkinson disease: A critical review of recent literature. *J Geriatr Psychiatry Neurol.* 2017;30(5):235–52. <https://pubmed.ncbi.nlm.nih.gov/28743212/>
83. Tuck KK, Brod L, Nutt J, Fromme EK. Preferences of patients with Parkinson's disease for communication about advanced care planning. *Am J Hospice Palliative Med.* 2015;32(1):68–77. <https://pubmed.ncbi.nlm.nih.gov/24052430/>
84. Ng JSC. Palliative care for Parkinson's disease. *Ann Palliat Med.* 2018;7(3):296–303. <https://pubmed.ncbi.nlm.nih.gov/29307212/>
85. Bouça-Machado R, Titova N, Chaudhuri KR, Bloem BR, Ferreira JJ. Palliative care for patients and families with Parkinson's disease. *Int Rev Neurobiol.* 2017;132:475–509. <https://pubmed.ncbi.nlm.nih.gov/28554419/>
86. Hasson F, Kernohan WG, McLaughlin M, Waldron M, McLaughlin D, Chambers H et al. An exploration into the palliative and end-of-life experiences of carers of people with Parkinson's disease. *Palliative Med.* 2010;24(7):731–6. <https://www.semanticscholar.org/paper/An-exploration-into-the-palliative-and-end-of-life-Hasson-Kernohan/8e8712d4db77e1d8e967fcd2515f0fd7de06b689>
87. Sudore RL, Lum HD, You JJ, Hanson LC, Meier DE, Pantilat SZ et al. Defining advance care planning for adults: A consensus definition from a multidisciplinary Delphi panel. *J Pain Symptom Manage.* 2017;53(5):821–32. <https://pubmed.ncbi.nlm.nih.gov/28062339/>
88. Sokol LL, Young MJ, Paparian J, Kluger BM, Lum HD, Besbris J et al. Advance care planning in Parkinson's disease: Ethical challenges and future directions. *NPJ Parkinsons Dis.* 2019;5:24. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6874532/>
89. The role of the health sector in the strategic approach to international chemicals management towards the 2020 goal and beyond. World Health Assembly resolution WHA69.4. Geneva: World Health Organization; 2016. <https://apps.who.int/iris/handle/10665/252784>
90. Preventing disease through healthy environments: A global assessment of the burden of disease from environmental risks. Geneva: World Health Organization; 2016. <https://apps.who.int/iris/handle/10665/204585>
91. WHO global strategy on health, environment and climate change: The transformation needed to improve lives and wellbeing sustainably through healthy environments. Geneva: World Health Organization; 2020. <https://apps.who.int/iris/handle/10665/331959>
92. iSupport for dementia. Training and support manual for carers of people with dementia. Geneva: World Health Organization; 2019. Licence: CC BY-NC-SA 3.0 IGO. <https://apps.who.int/iris/handle/10665/324794>
93. hGAP intervention guide for mental, neurological and substance use disorders in non-specialized health settings: mental health Gap Action Programme (mhGAP) – version 2.0. <https://apps.who.int/iris/bitstream/handle/10665/250239/9789241549790-eng.pdf?sequence=1>



**An urgent public
health response
is necessary
to meet the
health and social
requirements
of people with
Parkinson disease
and to improve
functioning,
quality of life and
prevent disability,
in the context
of providing
universal health
coverage.**



For more information please contact:

Brain Health Unit

Department of Mental Health and Substance Use

World Health Organization

Avenue Appia 20
CH-1211 Geneva 27
Switzerland

Email: brainhealth@who.int

Website: <https://www.who.int/health-topics/brain-health>

